

Credit Risk Prediction - Technical Report

GenAI Capstone Project - NST Sonipat - February 2026

1. Problem Statement

Access to credit is a fundamental enabler of economic participation, yet lending institutions face the persistent challenge of distinguishing creditworthy borrowers from those likely to default. Inaccurate risk assessments lead to significant financial losses through non-performing assets or the denial of credit to deserving applicants.

This project addresses the binary classification problem of predicting loan default risk. Given applicant attributes income, employment history, loan grade, prior default records - the goal is to predict whether a borrower will default (status=1) or repay (status=0).

The trained model is deployed as an interactive Streamlit web application for real-time credit risk evaluation.

2. Data Description

The dataset contains 32,581 records of historical loan applications.

Features

Feature	Type	Description
person_age	Numerical	Age of the applicant
person_income	Numerical	Annual income
person_emp_length	Numerical	Employment length (years)
person_home_ownership	Categorical	RENT, OWN, MORTGAGE, OTHER
loan_intent	Categorical	PERSONAL, EDUCATION, MEDICAL, etc.
loan_grade	Ordinal	Credit grade (A-G)
loan_amnt	Numerical	Loan amount requested
loan_int_rate	Numerical	Interest rate (%)
loan_percent_income	Numerical	Loan amount as % of income
cb_person_default	Binary	Previous default on record (Y/N)
cb_person_cred_hist	Numerical	Credit history length (years)
loan_status (Target)	Binary	0 = Non-Default, 1 = Default

Class Distribution: Non-Default ~78% | Default ~22%

3. Exploratory Data Analysis (EDA)

- Target Distribution: Confirmed class imbalance; addressed via balanced weighting.
- Income vs. Default: Defaulters tend to have lower incomes.
- Loan-to-Income Ratio: Strongest class separator; high ratios = higher risk.
- Interest Rate: Defaulters concentrated in higher rate brackets.
- Loan Grade: Default rates increase progressively from A to G.
- Correlation: loan_percent_income has strongest positive correlation with default.

4. Methodology

4.1 Preprocessing

1. Missing values: Median imputation (emp_length, int_rate)
2. Ordinal encoding: Loan grade A-G mapped to 1-7
3. Binary encoding: Default history Y/N to 1/0
4. One-hot encoding: Home ownership, loan intent (drop_first)
5. Feature scaling: StandardScaler normalization

4.2 Train-Test Split

75% training / 25% testing (random_state=42)

4.3 Models Trained

Model 1: Logistic Regression (baseline, max_iter=1000)

Model 2: Decision Tree Classifier (selected) - max_depth=10, min_samples_split=20, min_samples_leaf=10, class_weight='balanced'

Rationale: Decision Tree selected for higher accuracy (90.8% vs 84.5%) and better recall on the default class (0.77), which is critical in credit risk assessment.

5. Evaluation

5.1 Model Comparison

Metric	Logistic Regression	Decision Tree
Accuracy	84.52%	90.81%
ROC-AUC	0.8636	0.6817

5.2 Classification Report (Decision Tree)

Class	Precision	Recall	F1-Score	Support
Non-Default (0)	0.93	0.95	0.94	6331
Default (1)	0.81	0.77	0.79	1815
Weighted Avg	0.91	0.91	0.91	8146

5.3 Confusion Matrix

Actual / Predicted	Non-Default	Default
Non-Default	6,006 (TN)	325 (FP)
Default	424 (FN)	1,391 (TP)

6. Optimization & Limitations

- Class Imbalance: Used class_weight='balanced' to upweight minority class.
- Tree Pruning: max_depth=10 prevents overfitting while maintaining generalization.
- Limitation: Binary default history; cannot capture partial default rates.
- Limitation: ROC-AUC of 0.68 indicates room for probability calibration improvement.

7. Deployment

Live App: <https://genaicapstone-a7eipdbqudn2niewt9s2mp.streamlit.app/>

- Real-time prediction via interactive form.

- Three-tier decision: Approved (<20%), Needs Review (20-50%), Rejected (>50%).
- Displays default probability, repayment likelihood, and loan grade.

8. Team Contribution

Member	Contribution
Palak	Data preprocessing, EDA, model training & evaluation
Samarth	Streamlit app, UI/UX design, deployment, bug fixes
Jashvitha	Video presentation

9. Tech Stack

Component	Technology
Language	Python 3.x
ML Libraries	scikit-learn, pandas, NumPy
Visualization	Matplotlib, Seaborn
Web Framework	Streamlit
Deployment	Streamlit Cloud
Version Control	Git & GitHub

10. System Architecture

The system is structured into three functional layers:

- Layer 1 - User Interface (Streamlit): Handles applicant forms and input validation.
- Layer 2 - Intelligence (Python/Scikit-Learn): Manages preprocessing, scaling, and Decision Tree model logic.
- Layer 3 - Output Display: Visualizes risk metrics, tier status, and probability scores.

The Data Journey:

1. Input: User provides details (Income, Age, etc.) via the web form.
2. Transformation: Data is cleaned and scaled for the model.
3. Prediction: Decision Tree calculates default probability.
4. Action: Result is categorized and displayed on the dashboard.

11. Input-Output Specification

System Inputs (Borrower Profile)

Category	Key Features	Purpose
Demographics	Age, Emp Length	Assess stability and life-stage
Financials	Income, Loan Amount	Calculate Loan-to-Income ratio
History	Default/Credit Hist	Factor in past behavior
Context	Home/Loan Purpose	Collateral type and context

System Outputs (Credit Decision)

Result Type	Output Detail	Description
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Score	Default Prob %	Statistical risk of non-repayment
Decision	Status Tier	Approved / Review / Rejected
Metrics	Repayment %	Confidence in repayment

12. Key Risk Drivers (Ranked by Impact)

The following features are the primary drivers of credit risk, ranked by impact:

- 1. Loan-to-Income Ratio: Single biggest predictor; high debt is a primary rejection trigger.
- 2. Interest Rate: Strong signal of pre-existing risk profiles assigned by lenders.
- 3. Loan Grade: Summary quality metric; Grades E-G carry heavy penalties.
- 4. Annual Income: Baseline affordability check; low income keeps risk high.

13. References

1. scikit-learn Documentation - <https://scikit-learn.org/stable/>
2. Streamlit Documentation - <https://docs.streamlit.io/>
3. Credit Risk Dataset - Hosted on GitHub repository
4. Decision Tree Classifier - Breiman, L. (1984). Classification and Regression Trees